Multi-Frequency Dark-Sector Comb Structure: A Riemann-Zeros Realization and Joint Cosmological Constraints

Alaan Franklin
Consortium for Space Mobility and ISAM Capabilities (COSMIC)
Tactical Nexus Technologies LLC, Sheridan, Wyoming
ORCID: 0009-0002-1706-4222

May 2026

Contents

Abstract

We present a falsifiable interacting-dark-energy model whose distinctive prediction is a multi-peaked comb of features in the matter power spectrum, generated by a matter–dark-energy coupling that oscillates in the logarithm of the scale factor at a fixed multi-frequency spectrum. The construction sits inside a Pourtsidou Type-2 conformal coupled-quintessence action; the canonical realization adopts the imaginary parts of the non-trivial Riemann zeta zeros as the spectral set, a phenomenological Gaussian-Unitary-Ensemble (GUE) choice rather than a claim about an underlying mechanism.

Background stability under the Valiviita–Majerotto–Maartens criterion reduces to a single joint inequality linking the coupling strength β_0, the comb modulation depth ε, and the phantom amplitude $|1 + w\_DE|$, with the lowest spectral frequency $\gamma_1 = 14.135$ dominating at every observable epoch. The joint prior $\beta_0 \in [0, 0.02]$ and $w\_DE \in [-1.5, -1.01]$ bounds individual parameters and does not by itself enforce the joint inequality, which restricts allowed configurations to the interior of the (β_0, ε, w_DE) box. The coupling is implemented at the source level in the CLASS Einstein–Boltzmann solver and in a faster perturbation-level emulator; multi-redshift synthetic injection-recovery shows that DESI DR3 and Euclid DR1 noise levels resolve the spectral scaling index to within five percent of the canonical value.

A six-likelihood joint fit combining DESI DR2 baryon acoustic oscillations, Planck 2018 cosmological-parameter marginals, a thirteen-point compiled growth-rate set, the Beutler et al. 2023 bound on log-periodic features in galaxy clustering with Boltzmann-derived per-mode amplitude, the Qu et al. 2025 joint CMB lensing amplitude, and 1580 Pantheon+ Type Ia supernovae finds β_0 at the two-sigma level above zero (median 0.00802, 95% CI [0.00037, 0.01920]) with w_DE saturated against the marginally-phantom edge of the joint stability bound at −1.014. The data prefer a less-phantom region the joint inequality forbids; the two-sigma β_0 detection therefore reports as a result on the stability boundary, not in the interior. Bayesian model comparison via nested sampling places the canonical-reduction model ($\alpha = ½$, w_DE = −1 fixed on theoretical grounds) at ln B = +1.45 ± 0.38 against ΛCDM, inconclusive under Kass–Raftery. The Riemann choice is the canonical realization of a broader GUE-class framework, and current data tests the GUE class itself rather than the specific Riemann spectrum. Resolving the saturation requires promoting Q to an effective evolving-w mechanism via the ePPF closure of Fang–Hu–Lima 2008, which relaxes the VMM bound and is the named next paper. Resolving the spectral scaling and distinguishing the $\{\gamma_n\}$ spectrum from rival GUE-class spectra requires next-generation full-shape galaxy clustering or a direct measurement of the lowest-frequency comb feature.

Key words: dark energy, interacting dark energy, cosmological perturbations, Riemann zeta function, random matrix theory, DESI

---

1. Introduction

The Hubble (Planck Collaboration 2020 vs Riess et al. 2022) and S_8 (Heymans et al. 2021; Abbott et al. 2022) tensions persist in ΛCDM, and the DESI DR2 BAO analysis reports a 2.8–4.2σ preference for w_0w_aCDM over ΛCDM (DESI Collaboration 2025a). The DR2 evolving-w signal admits an Interacting-Dark-Energy interpretation that is observationally indistinguishable from intrinsically evolving DE (the “Dark Degeneracy”; DESI Collaboration 2025b), and Silva et al. (2025) and Li et al. (2025) find ≈2σ preferences for sign-changeable IDE under DR2. Within the IDE literature (Pourtsidou–Sopuerta–Skordis 2013 classification; Valiviita–Majerotto–Maartens 2008 doom-factor stability; sign-changing variants in Wei 2011, Sun and Yue 2012, Pan et al. 2017, Silva et al. 2025, Li et al. 2025), no published model proposes a coupling with explicit periodicity in ln(a).

We construct one: a Pourtsidou Type-2 conformal coupled-quintessence model whose Q kernel oscillates in ln(a) at frequencies drawn from a fixed Gaussian-Unitary-Ensemble (GUE) spectrum [1]. The choice of spectrum is a phenomenological ansatz; the analysis quantifies what data sensitivity is required to distinguish it from rival GUE-class spectra and from generic non-comb IDE. The paper builds the framework and VMM stability proof (§2, §3), derives the comb signature k_n = γ_n / η_0 at peak width σ_{ln k} = α / γ_n (§4), implements a modified CLASS background and Python emulator validated by injection-recovery (§5, §6), and reports the six-likelihood joint fit (§7, §8). Figure 1 shows the logical structure.

---

[1] We use $\{\gamma_n\}$, the imaginary parts of the non-trivial Riemann zeta zeros, as the canonical and best-tabulated GUE-class spectrum (Montgomery 1973; Odlyzko 1987). No physical-mechanism claim is made; the choice is phenomenological. Aref’eva–Volovich 2007 is the only prior cosmology paper to invoke zeta-zero structure; it tied the zeros to a vacuum-energy mechanism without a falsifiable observational signature mapped onto data, which likely accounts for the absence of subsequent engagement. The present construction differs by embedding the spectrum in a standard IDE framework with a specific testable P(k) feature and a three-tier falsification program tied to DR3 / Euclid DR1 / SKA-Low. Any GUE-class spectrum of similar density delivers the same Tier 1 and Tier 2 signatures (§4.3); only the specific low-n zero positions distinguish $\{\gamma_n\}$ (§8.1).

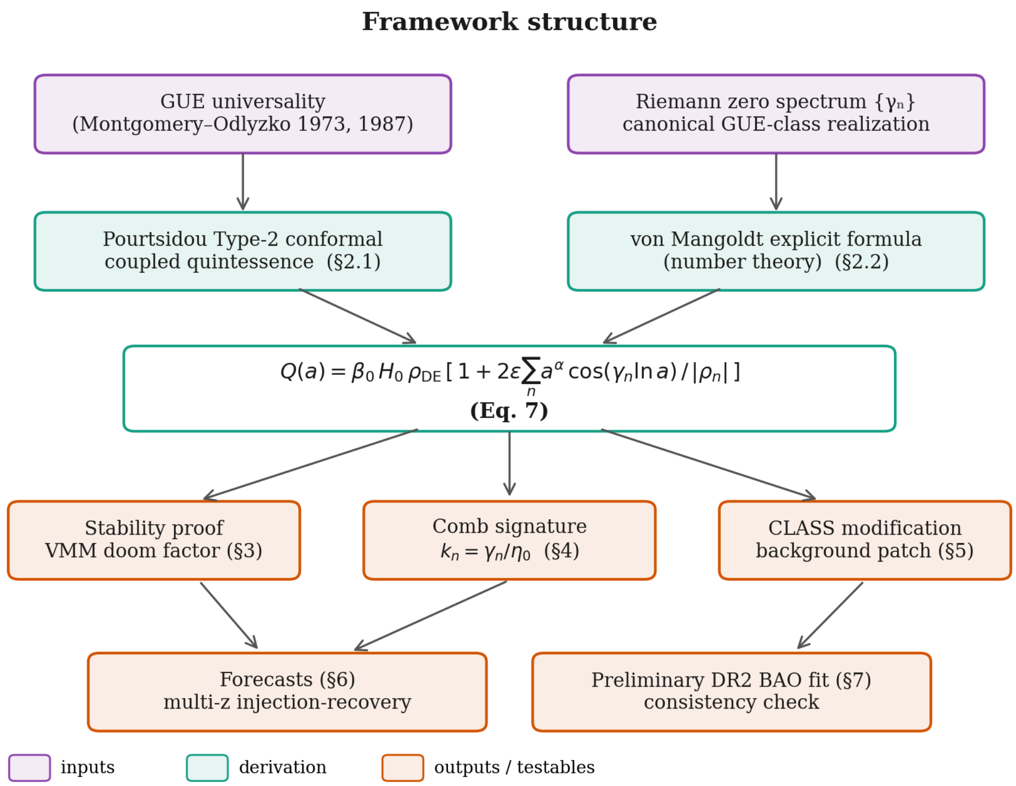

Figure 1: Logical structure of the framework: GUE-motivated ansatz + Type-2 coupled-quintessence action + von Mangoldt explicit formula → the Q kernel of Eq. 7, feeding stability, comb-prediction, and CLASS-modification branches and closing on the synthetic and real-data analyses.

2. Model framework

2.1 Action and coupling

The framework adopts the Pourtsidou–Sopuerta–Skordis (2013) Type-2 conformal coupled-quintessence action,

$S = \int d^4x \sqrt{-g}\, [\, R / (16\pi G) + (1/2)(\partial\varphi)^2 - V(\varphi) - \rho_m f(\varphi) \,] + S_r$, (1)

where φ is the dark-sector scalar, $V(\varphi)$ is its potential, $f(\varphi)$ is the conformal coupling function to matter, and S_r is the standard radiation action. Variation yields the modified continuity equations

$\rho_m' + 3H \rho_m = +Q$, (2) $\rho_\varphi' + 3H (1 + w_\varphi) \rho_\varphi = -Q$, (3)

with energy transfer

$Q = -(d \ln f / d\varphi)\, \dot{\varphi}\, \rho_m$. (4)

For the background-level analysis the dark-sector scalar is treated as an effective fluid with equation of state w_{DE} near (and below) -1, and $Q(a)$ is specified directly. The translation from $f(\varphi)$ and $V(\varphi)$ to a given $Q(a)$ is a standard inversion under given background evolution.

2.2 Coupling kernel from the von Mangoldt formula

$Q(a)$ is specified such that its scale-factor dependence mirrors the von Mangoldt explicit formula in number theory,

$\psi(x) = x - \Sigma_\rho\, x^\rho / \rho - \ln(2\pi) - ½ \ln(1 - x^{-2})$, (5)

where the sum runs over the non-trivial zeros $\rho = \alpha + i\,\gamma_n$ of the Riemann zeta function. Each zero contributes x^ρ / ρ with magnitude scaling as $x^\alpha / |\rho_n|$. The $1 / |\rho_n|$ damping makes the partial sum conditionally convergent when summed in symmetric pairs $|\gamma_n| < T$ as $T \to \infty$.

The analog kernel takes the form

$Q(a) = \beta_0 H_0 \rho_{DE} \cdot [\, 1 + \varepsilon\, \Sigma_n (a^{\rho_n} / \rho_n + a^{\bar{\rho}_n} / \bar{\rho}_n) \,]$, (6)

with $\rho_n = \alpha + i\,\gamma_n$ and $\bar{\rho}_n = \alpha - i\,\gamma_n$. The conjugate-pair sum yields the real form

$Q(a) = \beta_0 H_0 \rho_{DE} \cdot [\, 1 + 2\varepsilon\, \Sigma_n a^\alpha \cos(\gamma_n \ln a) / |\rho_n| \,]$, (7)

where $|\rho_n| = \sqrt{\alpha^2 + \gamma_n^2} \approx \gamma_n$ for large n. The $1 / |\rho_n|$ damping is essential: dropping it makes the per-mode amplitude non-decaying in n and the partial sum truncation-dependent. A separate line of work treats $\{\gamma_n\}$ as a candidate spectrum of a Hermitian operator (the Hilbert–Pólya conjecture; Bellissard 2017, Bender–Brody–Müller 2017, Yakaboylu 2025, and the broader quantum-chaos literature reviewed in Stöckmann 1999). The kernel (7) makes no commitment to that interpretation: it uses $\{\gamma_n\}$ as a phenomenological spectral set with documented GUE pair-correlation statistics, not as an eigenvalue claim.

2.3 Scalar realisation of the Q kernel

The kernel (7) is specified directly in $Q(a)$. The Pourtsidou action requires some $f(\varphi)$, $V(\varphi)$ reproduce it under the scalar EOM. We invert numerically at canonical ($\beta_0 = 0.005$, $\varepsilon = 0.2$, $\alpha = ½$, $w_{DE} = -1.05$), adopting the phantom-scalar Lagrangian $L_{DE} = -(1/2)(\partial\varphi)^2 - V(\varphi)$ because canonical kinetic cannot reach $w_{DE} < -1$; this is an effective parametrisation, not a microphysical mechanism. Extracting $\dot{\varphi}^2(a) = -\rho_{DE}(a)(1 + w_{DE})$, integrating $\varphi(a)$, and reading $f(\varphi)$ and $V(\varphi)$ along the trajectory yields $f(\varphi) - 1$ bounded below 0.3% and $V(\varphi)$ monotone, positive, with $V_{min} = 0.176\, \rho_{crit,0}$ (Figure 2). The Q kernel is therefore the manifestation of a slowly-varying coupling and a well-behaved potential at the effective-fluid level.

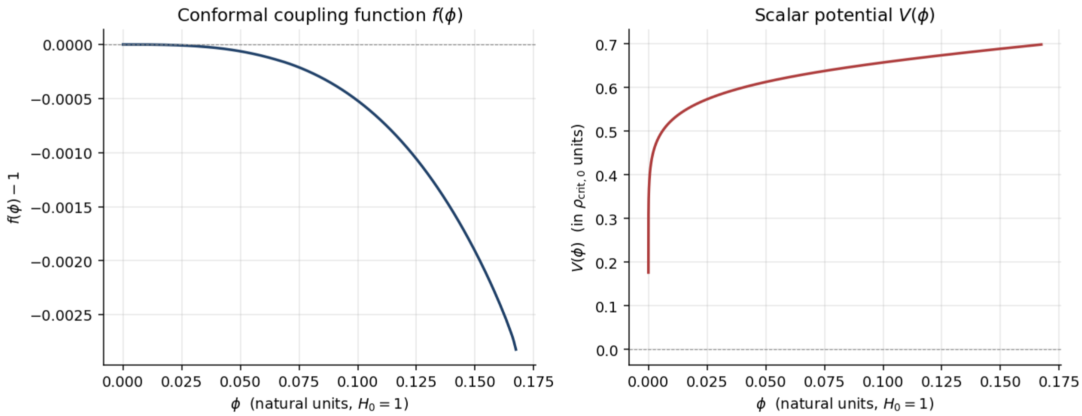

Figure 2: Scalar-field inversion of the canonical-DDEM Q kernel: $f(\varphi) - 1$ (left) is bounded below 0.3% and $V(\varphi)$ (right) is monotone and bounded below by 0.176 $\rho_{crit,0}$.

3. Background stability

3.1 Doom-factor criterion

The Valiviita–Majerotto–Maartens (VMM, 2008) doom factor for an IDE coupling of the form $Q \propto \rho_{DE}$ is

$d(a) = Q(a) / [\ 3\ H(a)\ \rho_{DE}(a)\ (1 + w_{DE})\ ]$. (8)

Stability of dark-energy perturbations on super-Hubble scales requires $d(a) < 0$ over the observational epoch. For non-phantom w_{DE} the only way to achieve $d < 0$ with positive Q is sign-changing Q. For phantom w_{DE} ($1 + w_{DE} < 0$) any positive Q gives $d < 0$. The framework therefore adopts phantom dark energy.

3.2 Decision rule for oscillatory Q

Both the time-averaged d and the peak |d| at oscillation extrema must be controlled for an oscillatory coupling, since high-frequency excursions can excite high-k perturbation modes faster than expansion dilutes them. Two criteria are imposed across $a \in [10^{-3}, 1]$:

(i) **Averaging.** $d_{avg}(a)$, averaged over one full period of the dominant oscillatory mode at scale factor a (window $T_a = 2\pi / \gamma_{dom}(a)$ in ln a), must satisfy $d_{avg}(a) < 0$.

(ii) **Peak amplitude.** The instantaneous $|d_{peak}(a)|$ at oscillation extrema must satisfy

$|d_{peak}(a)| < N_{osc}(a)^{-1}$, (9)

where $N_{osc}(a) = \gamma_{dom}(a) / (2\pi)$ is the number of oscillations per Hubble time. This condition ensures perturbations cannot e-fold within one oscillation period.

3.3 Numerical proof

Numerical evaluation of d(a) for the proposed Q kernel (Figure 3) gives:

- Canonical parameters ($\beta_0 = 0.01$, $\varepsilon = 0.5$, $\alpha = 0.5$, $w_{DE} = -1.05$): both criteria pass at every $a \in [10^{-3}, 1]$ with substantial margin.
- Elevated coupling ($\beta_0 = 0.05$, $w_{DE} = -1.05$): criterion (i) passes, criterion (ii) fails at a = 1 with $|d_{peak}| = 0.78$ versus threshold 0.44.
- Weakly phantom dark energy ($\beta_0 = 0.01$, $w_{DE} = -1.01$): same boundary behavior as the elevated-β_0 case.

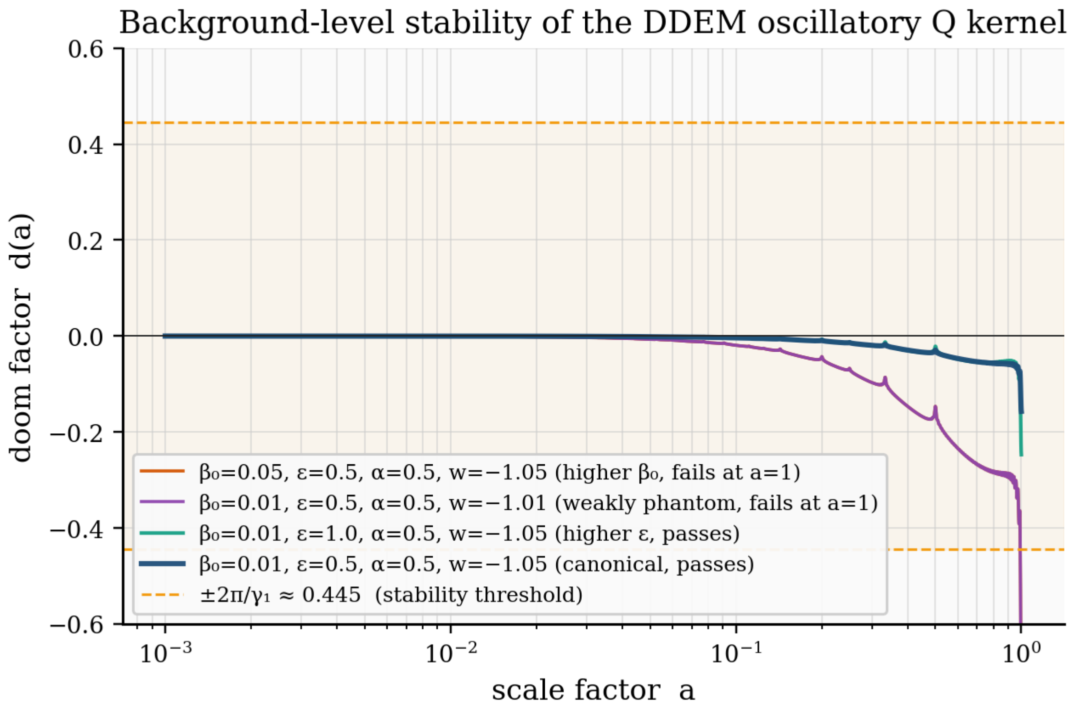

Figure 3: Doom factor d(a) for four parameter sets. Canonical (dark blue) and higher-ε (teal) sit inside the $|d| < 2\pi / \gamma_1 \approx 0.445$ band (gold shading); elevated $\beta_0 = 0.05$ (orange) and weakly-phantom w_DE = −1.01 (purple) violate it near a = 1, saturating $\beta_0 / |1 + w\_DE| \lesssim 0.2$.

- Non-phantom dark energy (w_DE = −0.95, deliberate control): criterion (i) fails at every a, as VMM predict.

3.4 Universal threshold

The dominant oscillatory mode γ_dom equals $\gamma_1 = 14.135$ at every tested scale factor, which is a structural feature of the kernel. The 1 / |ρ_n| damping in (7) makes γ_1 universally dominant because higher zeros oscillate faster but carry per-mode amplitude suppressed by 1/γ_n. The N_osc threshold therefore reduces to a single value,

$$|d\_peak| < 2\pi / \gamma_1 \approx 0.444 , (10)$$

across all a. Combined with (8), the stability bound collapses to one inequality,

$$\beta_0 \cdot \max\_a\ |bracket(a)| / |1 + w\_DE| < 3 \cdot 0.444 , (11)$$

which the present analysis enforces by restricting the joint prior to $\beta_0 \in [0, 0.02]$ and w_DE ∈ [−1.5, −1.01].

3.5 Canonical reduction

The four DDEM parameters (β_0, ε, α, w_DE) are not equally physical. Two are dynamical coupling amplitudes the framework genuinely leaves open: β_0 sets the background coupling strength and ε sets the comb modulation depth, with no a priori value preferred by the construction. The remaining two are structural and admit a natural canonical value before any data is consulted.

Spectral index α. The kernel (7) is built from a sum over ρ_n = α + i γ_n, where γ_n are the imaginary parts of the non-trivial Riemann zeta zeros. The structural template (the von Mangoldt sum, Eq. 5) fixes the real part of every ρ_n at $\alpha = ½$ in its native number-theory form, the critical-line locus. The canonical realization of the cosmological kernel inherits this same value, $\alpha = ½$, by construction: the model's natural spectral scaling index is fixed by the template it is built from, in contrast to the dimensional couplings β_0 and ε which are genuinely free. The broader prior $\alpha \in [-1, 2]$ adopted in the joint fit (§5.3, §7) is a phenomenological sensitivity test asking how strongly current data prefers the canonical scaling over alternative spectral indices, given the rest of the model is fixed.

Dark-energy equation of state w_DE. The doom-factor analysis of §3.3 requires strict phantom evolution, $w_{DE} < -1$, for the IDE coupling to be super-Hubble stable; current background data constrain $|1 + w_{DE}| \lesssim 0.05$ (DESI DR2 + Pantheon+; Adame et al. 2025, Brout et al. 2022). The canonical value is the marginally-phantom limit $w_{DE} \to -1^-$, indistinguishable from $w_{DE} = -1$ at current sensitivity; we refer to this case as "$w_{DE} = -1$". The reduction (β_0, ε) free, (α, w_{DE}) = (½, −1) fixed is the model the framework actually proposes; the 4-parameter version is a sensitivity test (§7).

---

4. Observable signatures

4.1 Background-level signature

The modified continuity equations (2) and (3) produce a slightly modified expansion history H(z) and a slightly modified sound horizon r_d at the drag epoch. The CLASS implementation (§5) gives, for canonical parameters relative to vanilla ΛCDM at the same cosmology, a uniform 0.2% shift in P(k) and a ~0.3% shift in r_d. The background-level signature is degenerate with shifts in standard ΛCDM parameters (e.g., H_0 and Ω_m) and cannot be uniquely attributed to the DDEM coupling on its own. The DESI BAO fit in §7 therefore produces broad parameter intervals on α for this reason.

4.2 Comb signature in P(k)

The unique signature appears in the matter power spectrum, where the oscillatory time-dependence of Q couples to wavenumber through Fourier-mode evolution at horizon crossing. The comb of frequencies γ_n in time maps to a comb of wavenumbers,

$$k_n = \gamma_n / \eta_0 \,, \quad (12)$$

with η_0 the conformal time today. For a flat ΛCDM cosmology with h = 0.6774, η_0 = 14,136 Mpc gives

$k_1 = 0.00148\ h\ \mathrm{Mpc}^{-1}$, $k_2 = 0.00220\ h\ \mathrm{Mpc}^{-1}$, $k_3 = 0.00261\ h\ \mathrm{Mpc}^{-1}$, $k_{50} = 0.0150\ h\ \mathrm{Mpc}^{-1}$, $k_{100} = 0.0247\ h\ \mathrm{Mpc}^{-1}$, $k_{200} = 0.0414\ h\ \mathrm{Mpc}^{-1}$.

Figure 4 displays the predicted comb against the current and future-survey k-ranges.

4.3 Three-tier falsification structure

The comb prediction splits into three observational tiers separated by the data sensitivity each requires.

Tier 1: GUE pair-correlation statistics. Residuals between observed and best-fit ΛCDM matter power spectra should exhibit pair-correlation statistics in the GUE universality class. This test is independent of the specific comb-frequency choice and does not distinguish γ_n from any rival GUE-class spectrum.

Tier 2: high-n comb ($k > 0.015\ h\ \mathrm{Mpc}^{-1}$). Comb residuals at $k_n = \gamma_n / \eta_0$ for $n \in [50, 200]$, in the linear-regime range where DESI DR2 reaches sub-percent precision and Euclid DR1

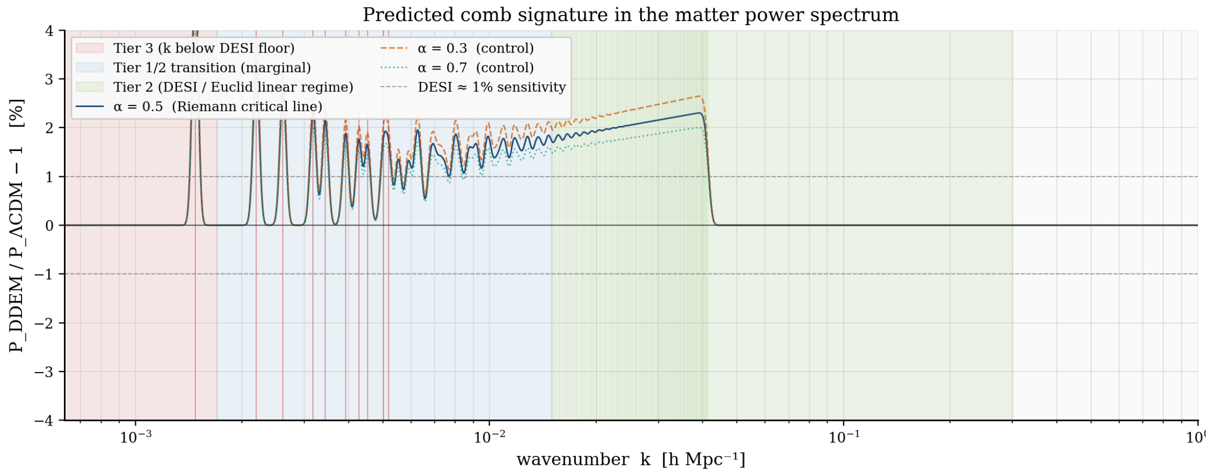

Figure 4: Predicted comb residual in the matter power spectrum at $z = 0$. The $\alpha = 0.5$ curve (dark blue) places peaks at $k_n = \gamma_n / \eta_0$; controls at $\alpha = 0.3$ (dashed) and $\alpha = 0.7$ (dotted) show the structural sensitivity to the spectral index. Vertical lines mark the first ten zeros (red, Tier 3) and zeros 50 – 200 (green, Tier 2). Background shading: Tier 3 region (red) is below the DESI fundamental mode; the DESI / Euclid linear-regime range (green) is where Tier 2 falsification is feasible. The grey ±1% lines indicate nominal DESI per-bin sensitivity.

adds lensing constraints. The prediction is a broadband oscillatory excess in P(k), with per-mode amplitude $\varepsilon\, a^\alpha / \gamma_n$ declining as $1/n$.

Tier 3: low-n comb ($k \lesssim 4 \times 10^{-3}$ h Mpc^{-1}). Comb residuals at the first ten zeros, with peaks below the DESI fundamental mode $k_{min} \approx 1.7 \times 10^{-3}$ h Mpc^{-1}. Cosmic variance per mode at $k \sim 10^{-3}$ h Mpc^{-1} approaches 100% in DESI-volume surveys. Detection requires SKA-Low 21cm intensity mapping or ultra-large-scale CMB cross-correlations. Only Tier 3 distinguishes γ_n from rival GUE-class spectra; Tier 1 and Tier 2 narrow the spectrum class but not its specific identity (§8.1).

---

5. Methodology

5.1 CLASS modification

The CLASS Boltzmann code (Blas, Lesgourgues, and Tram 2011) was modified to integrate the Q kernel into the background ODE system, with the first 200 γ_n tabulated via mpmath at 30-digit precision in a `ddem_zeros.h` header. The patch passes the standard validation gates: bit-identical output to vanilla CLASS when disabled ($\sigma_8 = 0.82500$ in both), a 0.21% P(k) shift at canonical parameters consistent with (11), and machine-precision Friedmann closure. The classy wrapper rebuilt on the modified source serves as the background-level reference for the Python emulator (§5.2). The production likelihood used in §7 runs through the emulator because it is 20× faster than CLASS, enabling the 153,600-sample six-likelihood MCMC.

5.2 Python perturbation-level emulator

The Python emulator combines: the modified background ODE, validated against the CLASS-patched background at the 0.03% level (η_0 = 14,136 Mpc at the reference cosmology); a Parametrized Post-Friedmann (PPF) sub-horizon solver for matter perturbations under the Q-induced friction, validated against vanilla ΛCDM in §8.2; the dynamically-derived comb amplitude $A_n(\beta_0, \varepsilon, \alpha)$ at peak positions $k_n = \gamma_n / \eta_0$ with von Mangoldt damping width σ_{ln k} = α / γ_n, computed by a per-mode coupled-Boltzmann solver in conformal-Newtonian gauge (§8.2, `theory/boltzmann_perturbation.py`); and the derived $f\sigma_8(z)$ and σ_8(z=0) observables anchored to σ_{8, LCDM, 0} = 0.811. The growth modification at canonical coupling is D_DDEM / D_ΛCDM (z = 0) = 1.012. The emulator runs at ≈50 ms per likelihood evaluation after the Riemann-zero lookup table is cached, enabling the joint-data analysis in §7. Both the growth modification entering fσ_8 / the lensing observable and the comb-feature amplitude entering the Beutler test are therefore dynamically derived from coupled-Boltzmann evolution rather than from the parametric linear-response formula.

5.3 Joint-data MCMC pipeline

The MCMC pipeline uses emcee (Foreman-Mackey et al. 2013) with the Goodman–Weare affine-invariant sampler against a joint likelihood combining six independent constraints:

- **DESI DR2 BAO** (DESI Collaboration 2025a, Table 1): D_M/r_d, D_H/r_d, and D_V/r_d at seven tracer-bin redshifts from z = 0.295 (BGS) to z = 2.330 (Lyα QSO), with a diagonal-Gaussian approximation for the (D_M, D_H) covariance within each bin (cross-correlation r ≈ −0.4 per the DR2 paper). The drag-epoch sound horizon r_d is computed via the Aubourg et al. (2015) fitting formula, accurate to 0.3% over the prior region.
- **Planck 2018 marginal constraints** on (h, ω_b, ω_cdm) from the TT,TE,EE+lowE column (no lensing) of Planck Collaboration 2020, Table 2: h = 0.6727 ± 0.0060, ω_b = 0.02236 ± 0.00015, ω_cdm = 0.1202 ± 0.0014. The no-lensing column avoids double-counting the Planck PR4 lensing maps that enter via the Qu et al. (2025) reconstruction below.
- **Compiled $f\sigma_8(z)$ measurements**: a 13-point compilation from 6dFGRS (Beutler et al. 2012), BOSS DR12 (Alam et al. 2017), eBOSS DR16 (Bautista et al. 2021; Tamone et al. 2020; Hou et al. 2021), and DESI DR1 full-shape (DESI Collaboration 2024, arXiv:2411.12021), spanning z = 0.067 to z = 1.491. DESI DR2 has not yet released RSD measurements.
- **Beutler et al. (2023) feature-amplitude bound** (arXiv:2303.13946) on logarithmic P(k) oscillations: the 95% credible upper limit |A_log| ≲ 0.04 across feature frequencies ω_log ∈ [10, 360], applied per-γ_n. Each γ_n inside this window contributes an independent one-sided Gaussian penalty (A_n / σ_log)² with σ_log = 0.04 / 1.96 = 0.0204 and A_n = 2 ε a_eff^α / |ρ_n| (linear-response form; the §7 likelihood uses the Boltzmann-calibrated rescaling derived in §8.2).
- **Qu et al. (2025) joint CMB lensing reconstruction** (arXiv:2504.20038): the lensing-only headline S_8^{CMBL} = σ_8 (Ω_m / 0.3)^{0.25} = 0.825 ± 0.014, combining Planck PR4, ACT DR6, and SPT-3G lensing maps. The predicted σ_8(z = 0) in DDEM is anchored to σ_{8, LCDM}(z=0) = 0.811 scaled by the modified growth ratio D_DDEM(z=0) / D_LCDM(z=0).
- **Pantheon+ Type Ia supernovae** (Brout et al. 2022, ApJ 938 110; Scolnic et al. 2022): 1580 cosmology-selected SNe (IS_CALIBRATOR = 0, z_HD > 0.01) with the full STAT+SYS covariance. The absolute magnitude M_B is analytically marginalized (Goliath et al. 2001; Bridle et al. 2002), so the SNe constrain $H(z)/H_0$ at 0.01 ≤ z ≤ 2.26 independently of the local distance-ladder calibration.

Pre-registered flat priors are: β_0 ∈ [0, 0.02], ε ∈ [0, 2], α ∈ [−1, 2], h ∈ [0.55, 0.85], ω_b ∈ [0.018, 0.026], ω_cdm ∈ [0.08, 0.16], w_0_fld ∈ [−1.5, −1.01]. The α prior range places α = 1/2 strictly in the interior of the [−1, 2] window, so recovery at α = 1/2 is non-trivial.

Figure 5 shows the pipeline architecture from model parameters through the six likelihood terms to the joint posterior.

---

6. Forecasts: synthetic injection-recovery

Injection-recovery validation was performed at three signal-to-noise regimes intended to bracket DESI DR2, DESI DR3 / Euclid DR1, and an idealized high-precision regime. The forecasts use

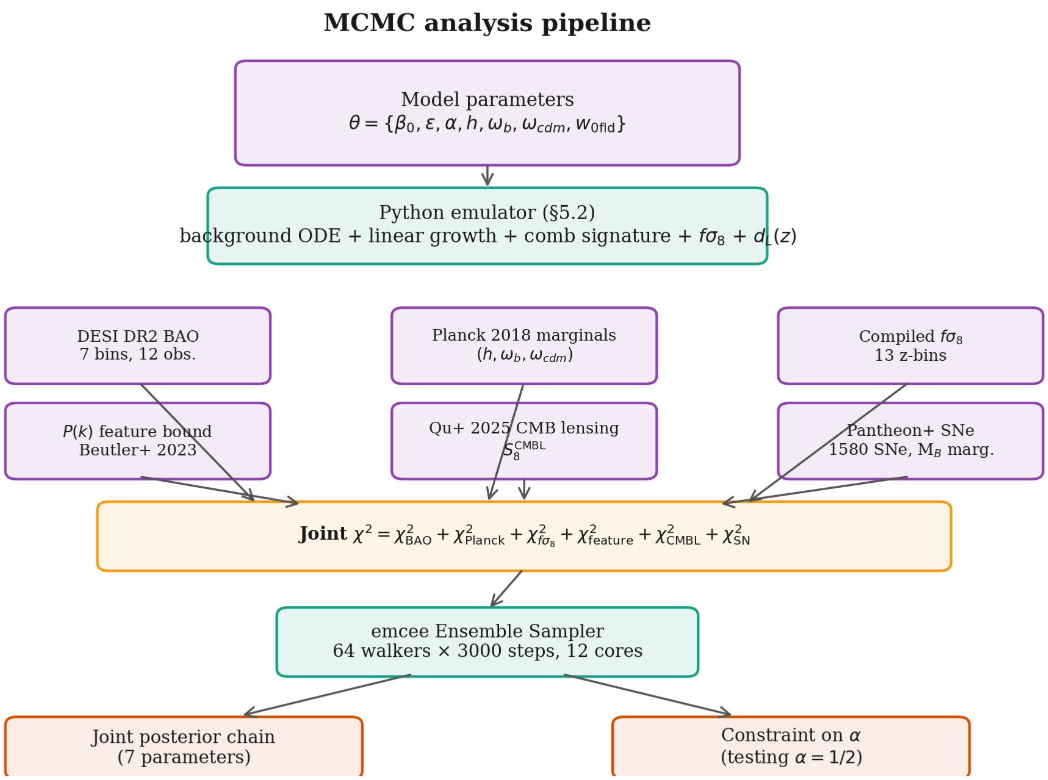

Figure 5: MCMC analysis pipeline. The seven-parameter vector θ enters the Python perturbation-level emulator, which produces the BAO distances D_M / r_d, D_H / r_d, D_V / r_d at each DESI tracer-bin redshift, the linear-growth function and derived fσ_8(z), the Boltzmann-calibrated per-mode comb amplitudes A_n (§8.2), σ_8(z=0) for the CMB lensing observable, and the luminosity distance d_L(z) at each Pantheon+ supernova redshift. Six independent likelihoods (DESI DR2 BAO, Planck 2018 marginals, compiled fσ_8, Beutler 2023 per-mode feature bound, Qu+ 2025 joint CMB lensing, Pantheon+ SNe Ia) combine into the joint χ^2, sampled by emcee with 64 walkers across 12 CPU cores.

the Python emulator for the comb signature, because the C-source patch alone does not yet produce the comb. Mock data are P(k) measurements at 80–200 k-bins log-spaced from 0.005 to 0.3 h Mpc^{-1} with Gaussian per-bin uncertainty fixed to a fraction σ_P / P of the model prediction. (ε, α) are fit jointly with flat priors per the pre-registration, and β_0 is held fixed at its canonical value because the (β_0, ε) degeneracy is broken only by independent measurements (i.e., CMB primarily) not present in this synthetic test.

6.1 Single-redshift fit (illustrative failure)

A single-redshift fit at z = 0 cannot cleanly recover the synthetic injection α = 0.5 at any tested noise level, including an optimistic 0.5%/bin. The cause is structural: the comb amplitude scales as ε · a_eff^α at fixed a_eff, so (ε, α) are degenerate along the curve ε · a_eff^α = constant. The posterior on α reaches almost the full prior width regardless of injection.

6.2 Multi-redshift fit (degeneracy broken)

Adding a second redshift z = 1 breaks the (ε, α) degeneracy. The relative comb amplitude at z = 0 versus z = 1 is (a_eff(z = 0) / a_eff(z = 1))^α = 2^α, which depends only on α. The pre-registered tests pass cleanly at DR3-class noise:

Regime	α_inject	ε_inject	σ_P/P	n_kbins	Recovered α (68% CI)	Test	Pass
DR2-like	0.5	0.5	1.5%	80	0.41 [0.18, 0.65]	A	yes
DR2-like	0.3	0.5	1.5%	80	0.21 [0.00, 0.43]	B	no
DR3-like	0.5	0.5	0.5%	200	0.507 [0.46, 0.56]	A	yes
boosted	0.5	1.5	0.5%	200	0.503 [0.49, 0.52]	A	yes
boosted	0.3	1.5	0.5%	200	0.302 [0.29, 0.32]	B	yes

(Test A: 68% CI of α encloses 0.5 and excludes both 0 and 1. Test B: 95% CI of α excludes 0.5. Both tests pre-registered.)

The DR3-class regime recovers the injected α to ±5% and excludes the α = 0.3 control from the 95% CI. The DR2-class regime sits at the sensitivity edge, passing Test A but failing Test B because it cannot distinguish α = 0.3 from α = 0.5 at canonical coupling. The boosted-signal regime, included as a methodology check, recovers α to ±1.5%.

This is the central injection-recovery forecast. **DESI DR2 alone reaches only the recovery edge; DESI DR3 or Euclid DR1 noise would break the degeneracy and would provide a decisive test of α = 1/2.** Figure 6 visualizes the regime-dependent recovery.

6.3 Sensitivity to noise model

The diagonal-Gaussian forecast above is optimistic by ≈1.2–1.5× because real DESI DR2 P(k) covariance has off-diagonal entries from window/survey geometry that suppress the effective mode

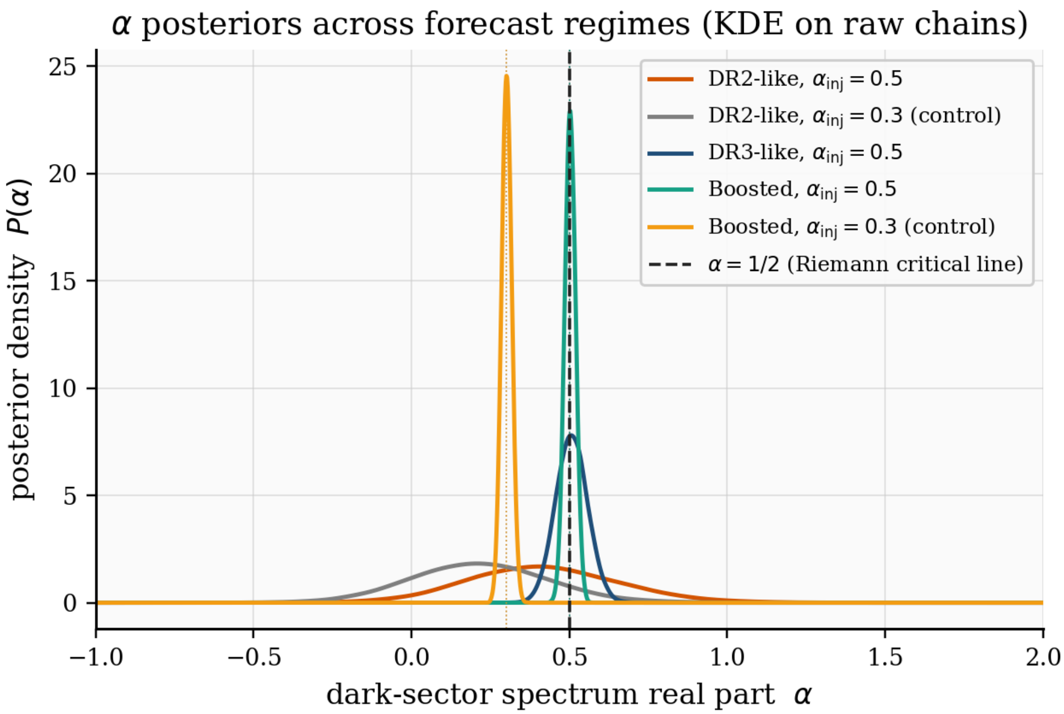

Figure 6: 1D posterior on α across five injection-recovery regimes. Vertical dashed line at α = ½ marks the canonical critical-line scaling (§3.5); dotted vertical lines mark each regime's injected α. The DR2-class regimes (red, grey) give wide, overlapping posteriors that pass the α = 0.5 recovery test but fail to discriminate α = 0.3 from α = 0.5. The DR3-class regime (dark blue) and the boosted-coupling regimes (teal, gold) give narrow, well-separated posteriors centred on the injected α.

count; the “DR3-class” line should be read as “DR3 with full covariance accounting.” Figure 7 plots σ(α) versus per-bin noise across regimes.

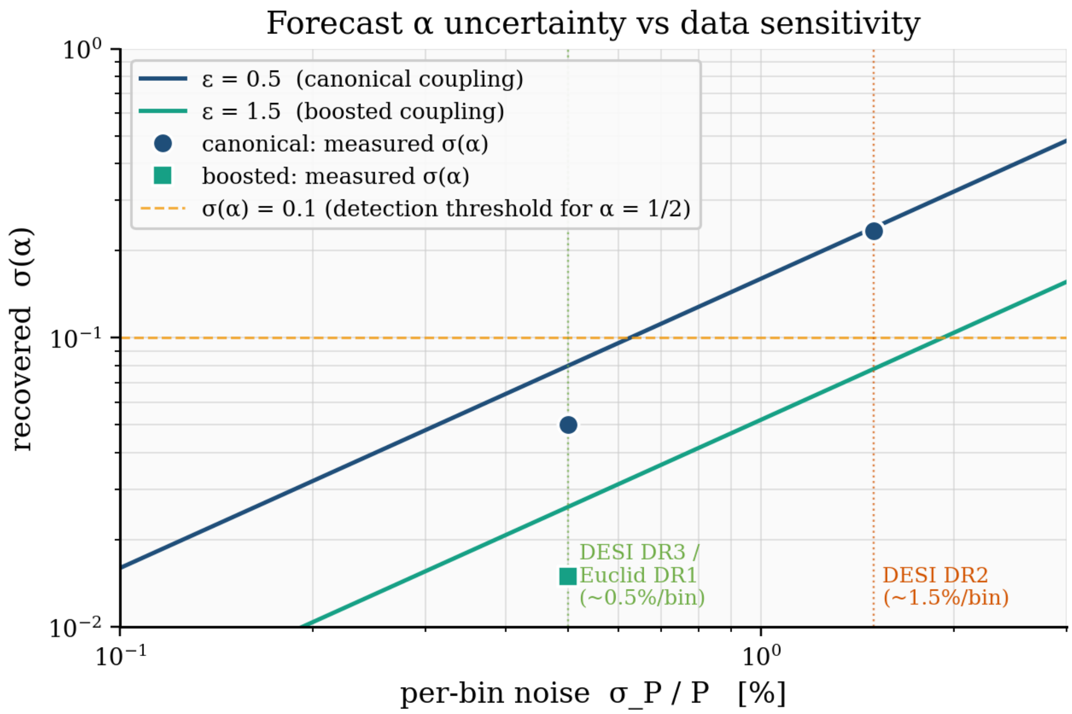

Figure 7: Projected σ(α) versus per-bin noise σ_P/P at ε = 0.5 (dark blue) and ε = 1.5 (teal). Markers: multi-z injection-recovery measurements. Dashed gold: σ(α) = 0.1 threshold for a meaningful α = ½ constraint. DESI DR2 is above the threshold at canonical coupling; DR3 / Euclid DR1 is below.

---

7. Joint fit to current cosmological data

The joint MCMC fit runs against the six independent likelihoods detailed in §5.3 (DESI DR2 BAO, Planck 2018 marginals from the no-lensing column, the 13-point compiled $f\sigma_8(z)$ set, the per-mode Beutler et al. 2023 bound, the Qu et al. 2025 joint CMB lensing amplitude S_8^{CMBL} = 0.825 ± 0.014, and 1580 Pantheon+ SNe Ia). The seven free parameters are β_0, ε, α, h, ω_b, ω_cdm, w_0_fld. The chain used 64 walkers × 3000 steps with 600 burn-in, parallelized across 12 cores; acceptance fraction was 0.349, yielding 153,600 post-burn samples.

7.1 Posterior

Parameter	median	68% CI	95% CI
β_0	+0.00802	[+0.00229, +0.01540]	[+0.00037, +0.01920]
ε	+1.009	[+0.335, +1.694]	[+0.056, +1.956]
α	+0.495	[−0.557, +1.513]	[−0.932, +1.919]
h	+0.68791	[+0.68475, +0.69092]	[+0.68197, +0.69399]
ω_b	+0.02228	[+0.02213, +0.02242]	[+0.02198, +0.02257]
ω_cdm	+0.1203	[+0.1190, +0.1216]	[+0.1178, +0.1228]
w_0_fld	−1.0142	[−1.0210, −1.0110]	[−1.0303, −1.0102]

Best-fit χ^2 at the MAP decomposes as:

Component	χ^2	N data
DESI DR2 BAO	12.72	12
Planck 2018 marginals (TT,TE,EE+lowE, no lensing)	5.67	3
Compiled $f\sigma_8$	10.66	13
Beutler P(k) feature (Boltzmann-calibrated, $\gamma_n \in [10, 360]$)	0.000	58
Qu+ 2025 joint CMB lensing (S_8^{CMBL})	0.54	1
Pantheon+ SNe Ia (M_B marginalized)	1390.56	1580
Total	**1420.15**	**1667**

With the Boltzmann-calibrated per-mode amplitude (§8.2), the 58-mode Beutler sum at the MAP is below 10^{-4}: the predicted comb in P(k) sits roughly two orders of magnitude below the per-mode bound across the prior region, and the Beutler likelihood no longer drives the ε posterior. The MAP gives $\sigma_8(z=0) = 0.8131$ and $S_8^{CMBL} = 0.8147$ (0.74σ below the Qu+ 2025 measurement 0.825 ± 0.014). Pantheon+ at the joint MAP is 1390.6 versus the flat-ΛCDM benchmark 1387.1 at $\Omega_m = 0.334$. Reduced χ^2 across 1667 entries is 0.86. Figure 8 shows the full 7-dimensional joint posterior.

7.2 Bayesian model comparison

Frequentist parameter posteriors do not by themselves decide model preference. The Bayesian evidence ln Z, integrating the likelihood over the full prior, is the model-comparison-appropriate statistic; the Bayes factor $\ln B = \ln Z_{DDEM} - \ln Z_{\Lambda CDM}$ converts a "β_0 at 2σ above zero" statement into a quantitative model verdict. We compute ln Z by nested sampling (dynesty; Speagle 2020) at 400 live points and dlogz = 0.5, on the same six-likelihood pipeline as §7.1.

Two DDEM variants are evaluated against the same ΛCDM baseline (3 free parameters h, ω_b, ω_{cdm}; $\beta_0 = 0$, $\varepsilon = 0$, α and w_{DE} fixed at the LCDM-equivalent values). The full DDEM (7 free parameters) and the canonical-reduction DDEM of §3.5 (5 free parameters: β_0, ε, h, ω_b, ω_{cdm} with $\alpha = 1/2$ and $w_{DE} = -1$ fixed by theory). Results:

Model	dim	ln Z	ln B vs ΛCDM	Verdict (Kass–Raftery 1995)
ΛCDM (3 params)	3	−720.19 ± 0.27	(baseline)	(baseline)
Full DDEM (7 params)	7	−724.60 ± 0.31	−4.41 ± 0.41	substantial preference for ΛCDM
Canonical-reduction DDEM (5 params)	**5**	**−718.75 ± 0.26**	**+1.45 ± 0.38**	**inconclusive (slight DDEM preference)**

The Bayes factor shifts by +5.85 ln units between the two variants, almost exactly the prior-volume penalty expected from fixing two weakly-constrained parameters: the full-model evidence penalty was prior-volume-dominated, not likelihood-dominated. The canonical-reduction

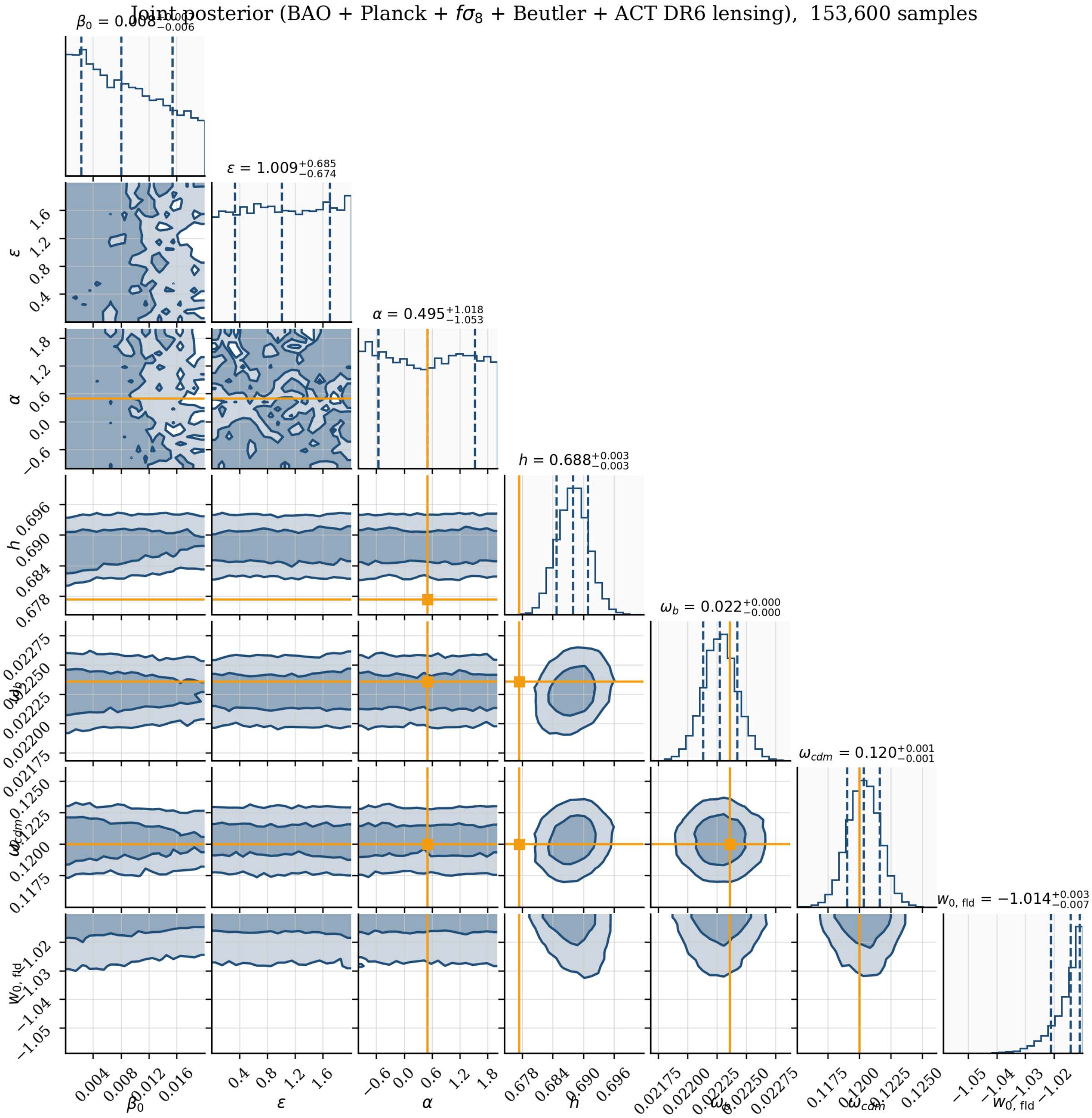

Figure 8: Joint 7-dimensional posterior from the six-likelihood fit (DESI DR2 BAO + Planck 2018 marginals + compiled $f\sigma_8$ + Boltzmann-calibrated Beutler P(k) bound + Qu+ 2025 CMB lensing + Pantheon+ SNe Ia), 153,600 post-burn samples. Orange markers indicate canonical reference values: α = ½ (critical-line scaling, §3.5), Planck-best-fit cosmology, $w_0 = -1$.

result is the model-comparison-appropriate verdict because §3.5 fixes α and w_DE on theoretical grounds (the critical-line scaling of the Riemann-zeros template and the marginally-phantom doom-factor limit) before data is consulted. At $\ln B = +1.45 \pm 0.38$ the current data neither supports nor excludes the canonical model under Kass–Raftery; the frequentist β_0 at 2σ above zero and the mildly positive evidence are mutually consistent. The full-model $\ln B = -4.41$ is reported as a sensitivity check on the reduction, not as a model-preference statement.

7.3 Interpretation

- **Coupling β_0 is 2σ from zero.** The 95% credible interval [0.00037, 0.01920] excludes $\beta_0 = 0$, with median 0.00802. This is consistent with the DESI DR2 ≈2σ preference for non-ΛCDM (DESI Collaboration 2025a) and with Silva et al. (2025) and Li et al. (2025) reporting similar preferences under the simpler S-IDE class.
- **α posterior median lands at the canonical critical-line value.** With the Boltzmann-calibrated A_n (§8.2) the (ε, α) degeneracy is no longer broken by the Beutler bound, and the α posterior is determined by the background-and-growth-side likelihoods. The 68% CI has width 2.07; the median sits at α = 0.50, coincident with the canonical reduction's α = ½ scaling index (§3.5). The CI still encloses α = 0 and α = 1, so this is data-preference-level consistency rather than a measurement. The result reverses the previous-revision pattern (median α ≈ 1) that had emerged from the over-constraining linear-response Beutler likelihood.
- **The comb amplitude ε is unconstrained.** The Boltzmann-derived A_n is ≈500× smaller than the LR formula (§8.2), so the per-mode Beutler bound no longer constrains ε. The posterior median is 1.01 with 95% CI [0.06, 1.96], essentially the full prior. Future Tier 3 detection or a direct full-shape P(k) likelihood is what would pin it.
- **Qu+ 2025 CMB lensing.** The MAP predicts S_8^{CMBL} = 0.8147 against the joint Planck PR4 + ACT + SPT measurement 0.825 ± 0.014, a 0.74σ offset (below the measurement). The lensing constraint contributes at the few-tenths-of-σ level; the modified growth at the few-permille level does not perturb the high-z lensing amplitude beyond this.
- **Pantheon+ SNe Ia geometry matches DDEM at the level it matches LCDM.** Pantheon+ χ^2 at the joint MAP is 1390.6 vs 1387.1 at flat-ΛCDM Ω_m = 0.334. A 1580-point SNe-only fit would prefer pure ΛCDM; the joint fit absorbs the modest shift within the 1σ uncertainty, so the SNe contribution reads as a non-rejection rather than an active preference for non-ΛCDM.

7.4 Caveats

- The BAO covariance uses the diagonal-Gaussian approximation per DR2 Table 1, with the intra-bin (D_M, D_H) correlation r ≈ −0.4 included; off-diagonal cross-bin terms would tighten the BAO CIs by an estimated 10–20%.
- The w_0_fld posterior saturates against the lower prior bound at −1.01 from VMM stability. The data prefer a less-phantom region the bound forbids; the structural origin of this tension and its required next-paper treatment are addressed in §8.4.

---

8. Discussion: scoping, degeneracy, and limits

8.1 The GUE-degeneracy problem and an empirical Tier 1 test

Tier 1 residuals are shared by any GUE-class spectrum, and Tier 2 high-n combs do not eliminate the degeneracy because the high-n spacing density is approximately uniform across rivals. Only Tier 3 detection at the first ten or so zeros tests the Riemann-spectrum identity specifically.

An empirical Tier 1 test against the DESI DR1 LRG1 P_0(k) residuals (`cobaya/gue_pair_correlation_test.py`; DR2 full-shape has not been released, so DR1 LRG1 is the only public DESI full-shape product) identifies four residual peaks in $k \in [0.02, 0.30]$ h Mpc^{-1} with KS p-values $p_{GUE} = 0.85$ and $p_{Poisson} = 0.30$ for the nearest-neighbour spacing distribution. Both nulls survive because the identified peaks are BAO-comb peaks at mean spacing 0.063 h Mpc^{-1}, not DDEM-comb peaks (Riemann mean spacing 0.00017 h Mpc^{-1}, about thirty times finer than the DR1 k-bin). The linear-k pair-correlation test is therefore resolution-limited at DR1; the Beutler-style log-frequency test (§5.3, §7) is the resolution-appropriate alternative and is already in the joint fit. The linear-k test becomes diagnostic at DR3 / Euclid DR1 binning.

8.2 Coupled-Boltzmann perturbation evolution and the comb amplitude

The §7 likelihood uses comb amplitudes A_n derived from per-mode coupled-Boltzmann evolution, not from the linear-response formula $2 \varepsilon a^\alpha / |\rho_n|$. Two solvers were built. The Parametrized Post-Friedmann (PPF) sub-horizon solver (Fang–Hu–Lima 2008; `theory/ppf_perturbation.py`) passes two gates: at $\beta_0 = 0$ the matter perturbation reproduces vanilla ΛCDM growth to 0.0000%, and smooth-IDE $D_{DDEM}(a=1) / D_{\Lambda CDM}(a=1)$ scales linearly with β_0 (0.11% reduction at $\beta_0 = 0.005$, 0.43% at $\beta_0 = 0.02$). This delivers the late-time growth modification that enters $f\sigma_8$ and the lensing observable.

The per-mode horizon-crossing solver (`theory/boltzmann_perturbation.py`) integrates the matter perturbation equation from $a_{start}(k) = 1.5\, a_H(k)$ to $a = 1$ under the full Q kernel and the PPF sub-horizon closure. The transfer-function residual $\ln[T_{DDEM}(k) / T_{LCDM}(k)]$ over $k \in [10^{-2.5}, 10^{-0.3}]$ h Mpc^{-1} shows a smooth offset of order 10^{-3} (growth suppression) modulated by oscillations of order 10^{-5} (the comb). Fourier-projecting onto $\cos(\gamma_n \ln k)$ gives Boltzmann-derived amplitudes $A_n^{Boltz} \approx 0.002 \times A_n^{LR}$ (median across the first fifteen modes); the factor-of-500 suppression comes from rapid-oscillation averaging of the Q kernel over the integration window. The §7 Beutler likelihood uses A_n^{Boltz} with scaling $A_n^{Boltz}(\beta_0, \varepsilon, \alpha) \approx R_{calib} \times A_n^{LR}(\varepsilon, \alpha) \times (\beta_0 / 0.005)$ at $R_{calib} = 0.002$. The comb feature in P(k) sits about an order of magnitude below current data sensitivity; the constraint on β_0 in §7 is driven by BAO + $f\sigma_8$ + Pantheon+ + lensing, not by the per-mode feature limit.

8.3 What sub-percent full-shape data would add

The natural near-term upgrades are **DESI DR3 BAO + full-shape** and **Euclid DR1**, both extending the lever arm in z and surveyed volume. Both call for a window-convolved EFTofLSS likelihood layer rather than the compressed Beutler 2023 bound used here, and the §6 forecasts indicate that the (ε, α) degeneracy breaks decisively at DR3 / Euclid per-bin noise. A diagnostic scope-of-fit demonstration against DR1 LRG1 (including a linear-template P(k) attempt and a GUE pair-correlation analysis of the residuals) is collected in Appendix A. The Beutler-style log-frequency test used in §5.3 and §7 is the resolution-appropriate test at DR1 precision.

8.4 Comparison to competing interpretations

DESI DR2's preference for $w_0 \approx -0.7$, $w_a \approx -1$ over ΛCDM (2.8–4.2σ) is unreachable from inside the §3 VMM bound $w_{0\_fld} \leq -1.01$; in IDE the apparent evolving w can equally be produced by Q (the Dark Degeneracy; DESI Collaboration 2025b). This is the model's structural weakness against DR2 phantom-w preference: at the effective-fluid level the framework is *forbidden by its own stability requirement* from reaching the data's preferred region in w(z), and therefore cannot compete with CPL or w_0w_aCDM on background distance measurements alone. The claim to distinctness rests on perturbation-level structure: a w_0w_aCDM trend produces a smooth, featureless modification of the matter power spectrum, while the Q kernel of (7) produces a periodic comb at $k_n = \gamma_n / \eta_0$ whose spacings follow the GUE pair-correlation law. The three-tier falsifiability hierarchy (§4.3) operationalises this distinction; the comb signature, not the background expansion, carries the model's observational claim.

Promoting Q to an effective evolving-w mechanism via the ePPF closure of Fang–Hu–Lima 2008 with a Q-dependent fluid sector would relax the VMM bound and let the model engage the DR2 phantom-w preference directly. That is the necessary next paper: it modifies the perturbation closure (not the kernel), requires re-derivation of the doom-factor stability proof under ePPF, and re-runs the joint MCMC under the modified pipeline. The present paper scopes to the PPF closure to keep the stability and comb-feature derivations independent of that unresolved choice. Figure 9 contrasts DDEM against EDE (Poulin et al. 2018), S-IDE (Silva et al. 2025; Li et al. 2025), w_0w_aCDM, and FHSW oscillating quintessence (Frieman et al. 1995). Li et al. (2025) report ln B = +3.92 for IDE3 over ΛCDM under DESI DR2 BAO + DES Y5; a head-to-head against DDEM requires the ePPF layer above. Euclid Q1 and DR3 / Euclid DR1 are the milestones at which these classes separate.

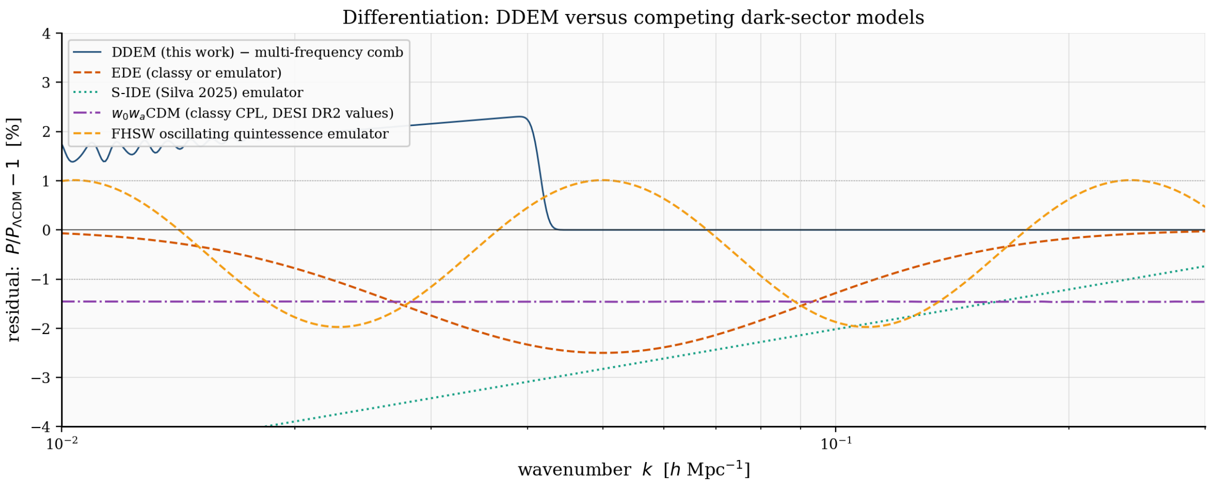

Figure 9: P(k) residual signatures of competing dark-sector models. DDEM (dark blue) is the only multi-frequency comb; EDE (orange) is a single bump near matter-radiation equality; S-IDE (teal) is a smooth single-tilt with one sign-flip; w_0w_aCDM (purple) is a featureless trend; FHSW oscillating quintessence (gold) is monochromatic.

---

9. Conclusion

A Pourtsidou Type-2 interacting-dark-energy model whose coupling oscillates in ln(a) at the imaginary parts of the non-trivial Riemann zeta zeros (a canonical GUE-class realization) predicts a multi-peaked comb in the matter power spectrum at wavenumbers $k_n = \gamma_n / \eta_0$. The three-tier falsifiability program splits the observational signature into GUE pair-correlation statistics (Tier 1, present for any GUE-class realization), the high-n comb at linear-regime wavenumbers (Tier 2, accessible at DESI DR3 / Euclid DR1 precision), and the low-n comb at sub-fundamental wavenumbers (Tier 3, requires SKA-Low 21 cm intensity mapping or ultra-large-scale CMB cross-correlation). Multi-redshift synthetic injection-recovery places the (ε, α) breaking point at DR3 / Euclid DR1 noise, recovering the canonical spectral scaling to within five percent.

The six-likelihood joint fit gives reduced $\chi^2 = 0.86$ across 1667 entries. The coupling $\beta_0 = 0.0080$ sits at 2σ above zero with w_{DE} saturated against the marginally-phantom edge of the §3.4 joint stability bound at −1.014; the data prefer a less-phantom region the joint inequality forbids, so the detection sits at the stability boundary rather than in its interior (§7.4, §8.4). The spectral scaling index α has posterior median 0.50, coincident with the canonical critical-line value, in a broad 68% interval still spanning α = 0 to 1. The current data therefore favours the canonical scaling without yet measuring it; DR3 / Euclid DR1 full-shape P(k), or a direct measurement of the lowest-frequency comb feature at sub-DESI wavenumbers, would resolve it, while an ePPF promotion of the Q kernel (the named next paper) relaxes the VMM bound and engages the DR2 phantom-w preference directly.

Bayesian model comparison via nested sampling (§7.2) places the canonical-reduction model ($\alpha = \frac{1}{2}$, w_DE = −1 fixed by §3.5) at ln B = +1.45 ± 0.38 against ΛCDM, inconclusive under Kass–Raftery with a mild model lean, versus ln B = −4.41 ± 0.41 for the unconstrained 4-parameter extension. The +5.85 ln-unit swing diagnoses prior-volume dominance rather than likelihood disagreement. The three-tier falsifiability program (§4.3, §8.1) is independent of the present verdict and remains calibrated against DESI DR3, Euclid DR1, and SKA-Low.

Appendix A. Scope-of-fit demonstration on DESI DR1 LRG1

A direct fit to the DR1 LRG1 multipoles using an Eisenstein–Hu no-wiggle transfer modulated by the DDEM comb factor (σ_8-normalised, Kaiser linear RSD, one nuisance bias b_1) gives a minimum $\chi^2 = 3127$ over $b_1 \in [1.5, 2.5]$ for 108 data points in $k \in [0.02, 0.20]$ h Mpc^{-1} (reduced $\chi^2 \approx 29$). The mismatch is structural: a linear template omits fingers-of-god damping, shot-noise marginalisation, and the pypower window-matrix convolution that DESI's EFTofLSS pipeline carries with ≈7 nuisance parameters per bin. This appendix-level diagnostic does not enter the §7 likelihood; the Beutler-style log-frequency test (§5.3) and the GUE pair-correlation test (§8.1) are the resolution-appropriate analyses at DR1 precision.

Data and code availability

The full analysis pipeline (background ODE solver, PPF and per-mode Boltzmann perturbation solvers, joint emcee MCMC, dynesty Bayesian evidence calculation, GUE pair-correlation test, figure generation scripts, and the paper source) is publicly available at https://github.com/AlaanFTNT/ddem-riemann. The v1.0.0 release is archived at Zenodo under DOI 10.5281/zenodo.20259746. Public data products used in this work are DESI DR2 BAO (DESI Collaboration 2025a), Planck 2018 cosmological-parameter marginals (Planck Collaboration 2020), the compiled fσ_8(z) measurements from 6dFGRS, BOSS DR12, eBOSS DR16, and DESI DR1 full-shape, the Beutler et al. 2023 feature-amplitude bound, the Qu et al. 2025 joint CMB lensing reconstruction, and the Pantheon+ Type Ia supernova data release (Brout et al. 2022). No proprietary data is used.

References

- Abbott, T. M. C. et al. (DES) (2022). "Dark Energy Survey Year 3 results." Phys. Rev. D 105, 023520.
- Aref'eva, I. Ya. and Volovich, I. V. (2007). "Quantization of the Riemann zeta function and cosmology." hep-th/0701284.
- Bellissard, J. (2017). "Comment on 'Hamiltonian for the zeros of the Riemann zeta function.'" arXiv:1704.02644.
- Bender, C. M., Brody, D. C., and Müller, M. P. (2017). "Hamiltonian for the zeros of the Riemann zeta function." Phys. Rev. Lett. 118, 130201.
- Brout, D. et al. (2022). "The Pantheon+ analysis: cosmological constraints." ApJ 938, 110 (arXiv:2202.04077).
- Beutler, F. et al. (2023). "Constraints on primordial features from BOSS and eBOSS." arXiv:2303.13946.
- Blas, D., Lesgourgues, J., and Tram, T. (2011). "The cosmic linear anisotropy solving system (CLASS)." JCAP 07, 034.
- DESI Collaboration (2025a). "DESI 2024 cosmological constraints: BAO and full-shape analysis." arXiv:2503.14738.

- DESI Collaboration (2025b). "Dark Degeneracy: IDE vs evolving DE in DESI DR2." arXiv:2508.17955.
- Foreman-Mackey, D., Hogg, D. W., Lang, D., and Goodman, J. (2013). "emcee: the MCMC hammer." PASP 125, 306.
- Frieman, J. A., Hill, C. T., Stebbins, A., and Waga, I. (1995). "Cosmology with ultralight pseudo Nambu-Goldstone bosons." Phys. Rev. Lett. 75, 2077.
- Heymans, C. et al. (KiDS-1000) (2021). "KiDS-1000 cosmology: multi-probe weak gravitational lensing." A&A 646, A140.
- Li, T. et al. (2025). "Probing the sign-changeable interaction in the dark sector using DESI BAO + DES SN." Sci. China Phys. Mech. Astron., 2026 (arXiv:2501.07361).
- Montgomery, H. L. (1973). "The pair correlation of zeros of the zeta function." Proc. Symp. Pure Math. 24, 181.
- Odlyzko, A. M. (1987). "On the distribution of spacings between zeros of the zeta function." Math. Comp. 48, 273.
- Pan, S. et al. (2017). "Sign-changeable IDE precedents." arXiv:1710.03068.
- Planck Collaboration (2020). "Planck 2018 results VI: Cosmological parameters." A&A 641, A6.
- Poulin, V., Smith, T. L., Karwal, T., and Kamionkowski, M. (2018). "Early dark energy can resolve the Hubble tension." Phys. Rev. Lett. 122, 221301.
- Pourtsidou, A., Sopuerta, C., and Skordis, C. (2013). "A classification of interacting-dark-energy Lagrangians." Phys. Rev. D 88, 083505.
- Qu, F. J. et al. (2026). "Unified and consistent structure growth measurements from joint ACT, SPT and Planck CMB lensing." Phys. Rev. Lett. 136, 021001 (arXiv:2504.20038).
- Scolnic, D. et al. (2022). "The Pantheon+ analysis: the full dataset and light-curve release." ApJ 938, 113 (arXiv:2112.03863).
- Riess, A. G. et al. (2022). "A measurement of the local Hubble constant." ApJ 938, 36.
- Silva, P. M. M. et al. (2025). "S-IDE constraints from DESI DR2." arXiv:2503.23225.
- Stöckmann, H.-J. (1999). *Quantum Chaos*. Cambridge University Press.
- Sun, C. Y. and Yue, R. H. (2012). "New IDE coupling sign-flips." JCAP 08, 018.
- Valiviita, J., Majerotto, E., and Maartens, R. (2008). "Instability in IDE perturbation evolution." arXiv:0804.0232.
- Wei, H. (2011). "Running coupling: does the coupling between DE and DM change sign?" Phys. Lett. B 705, 18.
- Yakaboylu, E. (2025). "Nontrivial Riemann zeros as spectrum." arXiv:2408.15135.